

Homework 3

Due at 11:59 pm, April 27th

1 Problem 1 (40 points): Comparing Gradient Descent and Newton's Method

Consider the quadratic optimization problem

$$\min_{x \in \mathbb{R}^2} f(x),$$

where

$$f(x) = \frac{1}{2} x^\top Q x - b^\top x, \quad Q = \begin{bmatrix} 10 & 0 \\ 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Use the same initial guess for **(2)**-**(6)**:

$$x^{(0)} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Unless otherwise stated, all iterations and plots should start from this point.

(1) Basic analysis

- (a) Show that $f(x)$ is strongly convex.
- (b) Compute the gradient $\nabla f(x)$ and Hessian $\nabla^2 f(x)$.
- (c) Find the unique minimizer x^* analytically.

(2) Consider gradient descent with fixed step size

$$x^{(k+1)} = x^{(k)} - \alpha \nabla f(x^{(k)}),$$

using $\alpha = 1$.

- (a) Write out the iteration explicitly for this problem.
- (b) Compute the first two iterations by hand.
- (c) Based on your calculations, discuss whether this method converges from the given initial condition.

(3) Now consider steepest descent with exact line search:

$$x^{(k+1)} = x^{(k)} - \alpha_k \nabla f(x^{(k)}),$$

where

$$\alpha_k = \arg \min_{\alpha \geq 0} f\left(x^{(k)} - \alpha \nabla f(x^{(k)})\right).$$

- (a) Derive a closed-form expression for α_k .
- (b) Compute the first two iterations.

- (c) Compare your results with those in Part 2.
- (4) Use gradient descent with backtracking line search. At each iteration, choose the search direction

$$p_k = -\nabla f(x^{(k)}),$$

and determine the step size α_k by backtracking until the first Armijo condition is satisfied:

$$f(x^{(k)} + \alpha_k p_k) \leq f(x^{(k)}) + c \alpha_k \nabla f(x^{(k)})^\top p_k.$$

Use the following parameters:

$$\alpha_0 = 1, \quad \rho = 0.5, \quad c = 10^{-4}.$$

- (a) Implement the method.
 - (b) Report the step sizes selected during the first two iterations.
 - (c) Compare the convergence behavior with exact line search.
- (5) Apply Newton's method:
- $$x^{(k+1)} = x^{(k)} - (\nabla^2 f(x^{(k)}))^{-1} \nabla f(x^{(k)}).$$
- (a) Write out the Newton step explicitly for this problem.
 - (b) Compute the first iterate from $x^{(0)}$.
 - (c) Explain why Newton's method behaves the way it does for this problem.
- (6) Numerical comparison. Implement all four methods in Matlab or Paython (for 3 iterations other than Newton's method):
- (a) gradient descent with unit step size,
 - (b) gradient descent with exact line search,
 - (c) gradient descent with approximate line search,
 - (d) Newton's method.

For each method:

- (a) Plot the trajectory of the iterates on the contour plot of $f(x)$.
- (b) Plot $f(x^{(k)}) - f(x^*)$ versus iteration number.
- (c) Plot $\|\nabla f(x^{(k)})\|$ versus iteration number.

Then answer the following questions:

- (a) Which method converges fastest in terms of iteration count?
- (b) How does the condition number of Q affect the performance of gradient descent?

Submission instructions

Please submit:

1. your derivations for Parts 1–5,
2. your code (you can incorporate your code in your submission instead of submitting a separate document),
3. your plots for Part 6,
4. a brief discussion comparing the methods.

2 Problem 2 (60 points): Unconstrained LQ-MPC Example

Consider an unconstrained LQ-MPC problem for a system with a discrete-time model of the form

$$x_{k+1} = Ax_k + Bu_k,$$

where

$$A = \begin{bmatrix} \frac{4}{3} & -\frac{2}{3} \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Let

$$C = \begin{bmatrix} -\frac{2}{3} & 1 \end{bmatrix},$$

and define the weighting matrices as

$$Q = C^\top C, \quad R = 0.001.$$

- (1) Is the open-loop system stable? Is (A, B) controllable? Is (A, C) observable?
- (2) For LTI discrete-time systems, the transfer function is given by

$$T(z) = C(zI - A)^{-1}B + D,$$

where the variable z plays a role similar to the Laplace variable s in continuous time. Use symbolic computations,

```
z = sym('z')
simplify(C*inv(z*eye(2,2) - A)*B + 0)
```

to compute the transfer function. Where are the poles (zeros of the denominator) and zeros (zeros of the numerator) of the open-loop transfer function? Are they inside or outside the unit disk of the complex plane? Are the poles the same as the eigenvalues of A ? Read more in the literature about the z -transform and discrete-time transfer functions.

- (3) Generate the step response of the open-loop system. Since our model is discrete-time, use the `dstep(A,B,C,0)` command rather than the `step` command. Note that the step response shows an initial undershoot, which is a symptom of a nonminimum-phase (unstable) zero.
- (4) Implement a Matlab function that performs unconstrained LQ-MPC computations shown in class. Use a template

```
[S, M, Qbar, Rbar, KON] = uncMPC(N, A, B, Q, R, P)
```

Your function should implement all the computations manually and should not call any MPC toolboxes or QP solvers. You can use either the batch (direct) approach or the recursive (DP) approach.

- (5) For the prediction horizon N between 2 and 20, use your function in part (d) and compute the feedback gain $K_{0,N}$. Generate a plot of the spectral radius of the closed-loop dynamics matrix

$$A + BK_{0,N}$$

versus the horizon N . For what values of the horizon N is the closed-loop stable?

- (6) Generate now the LQR gain K_∞ using the `dlqr.m` command. Use the sign convention from class (multiply the gain computed by `dlqr.m` by -1). Give and compare the values of $K_{0,20}$ and K_∞ . Are they close? Is the LQR gain stabilizing?
- (7) Generate a plot of each of the two components of the gain $K_{0,N}$ versus the horizon N . Indicate the values of K_∞ on the same plot by dashed horizontal lines. Does $K_{0,N}$ converge to K_∞ as $N \rightarrow \infty$ based on your plot?
- (8) Summarize your key takeaways from this example.